Rotations & Fictitious Forces

1. A block of mass m rests on a smooth incline of angle θ inside an elevator accelerating upward with constant acceleration a. The block is attached to the incline by a light spring parallel to the plane. After equilibrium is reached, what is the spring extension?

   A. $\dfrac{mg\sin\theta}{k}$

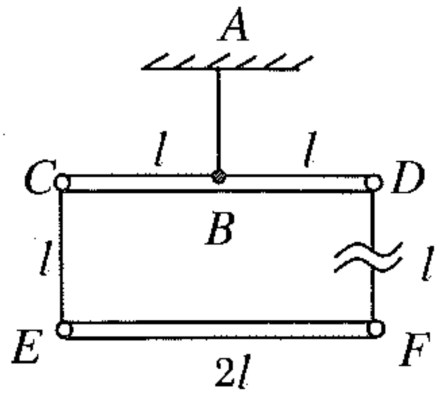
   B. $\dfrac{m(g+a)\sin\theta}{k}$

   C. $\dfrac{m(g-a)\sin\theta}{k}$

   D. $\dfrac{ma\sin\theta}{k}$

   E. $\dfrac{m\sqrt{g^2+a^2}\sin\theta}{k}$

2. Two light rigid rods CD and EF, each of length 2ℓ, lie in a vertical plane. A point mass m is fixed at each end of both rods. Points C and E, and points D and F, are connected by two inextensible light strings CE and DF, each of length ℓ. The midpoint B of rod CD is suspended from the ceiling at point A by a light string AB. Initially the system is at rest. The string DF is suddenly cut. Immediately after the cut, what is the tension in string AB?



   A. $\dfrac{4}{3}mg$

   B. $2mg$

   C. $\dfrac{8}{3}mg$

   D. $3mg$

   E. $4mg$

3. In the setup of the previous problem, suppose the rods each uniformly have mass m instea, and there are no longer point masses on their ends. What is the tension in string AB now immediately after the cut?

   A. mg

   B. $\dfrac{6}{7}mg$

   C. $\dfrac{8}{7}mg$

   D. $\dfrac{4}{3}mg$

   E. $2mg$

4. A block rests on a smooth wedge of angle θ. The wedge is accelerated horizontally to the right with acceleration a. Find the condition on a such that the block does not slip relative to the wedge.

   A. $a = g\sin\theta$

   B. $a = g\cos\theta$

   C. $a = g\tan\theta$

   D. $a = g\cot\theta$

   E. $a = \sqrt{2}g\tan\theta$

5. A uniform rod of length L and mass M is pivoted at one end. It is released from rest from a horizontal position. What is the angular speed of the rod after it has rotated downward by an angle θ?

   A. $\sqrt{\dfrac{3g}{L}\sin\theta}$

   B. $\sqrt{\dfrac{2g}{L}\sin\theta}$

   C. $\sqrt{\dfrac{3g}{2L}\sin\theta}$

   D. $\sqrt{\dfrac{g}{L}(1-\cos\theta)}$

   E. $\sqrt{\dfrac{2g}{L}(1-\cos\theta)}$

6. A point mass m is attached to one end of a light, inextensible string of length ℓ. The other end of the string is attached to a very thin, vertical, frictionless pole. Uniform gravitational acceleration g acts downward.

   Initially, the string is wound tightly around the pole in a very large number of small horizontal turns, and the mass is touching the pole. The mass is released from rest, and the string gradually unwinds while remaining taut. When the string has just completely unwound from the pole, what angle θ does the string make with the vertical pole?

   A. $\arctan(1)$

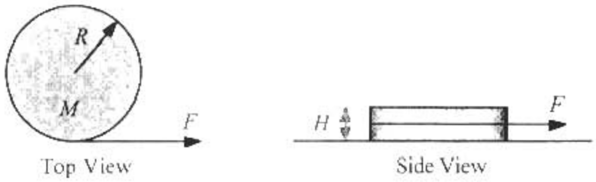
   B. $\arctan(\sqrt{2})$

   C. $\arcsin\left(\sqrt{\dfrac{2}{3}}\right)$

   D. $\arccos\left(\dfrac{1}{\sqrt{3}}\right)$

   E. $\arccos\left(\dfrac{1}{2}\right)$

7. A thin uniform disk of mass M and radius R rests on a frictionless horizontal table. A long, light, inextensible cord is wrapped tightly around the rim of the disk and is pulled horizontally with a constant force F. The disk starts from rest and both translates and rotates as the cord unwinds. Neglect any slipping between the cord and the disk.



What is the ratio of the disk's rotational kinetic energy to its translational kinetic energy?

A. $\frac{1}{2}$

B. 1

C. 2

D. 3/2

E. 8/3

8. A coin of radius r rolls without slipping around the *outside* of a fixed coin of radius $2r$ in the same plane. After one full circuit around the fixed coin, how many full revolutions does the rolling coin make about its own center?

A. 2

B. 3/2

C. 3

D. 7/2

E. 4

9. A coin of radius r_2 rolls without slipping along the *inside* of a fixed, thin circular loop of radius r_1, where $r_2 < r_1$. After one full circuit inside the loop, how many full revolutions does the coin make about its own center? (Count revolutions in magnitude.)

A. $\frac{r_1}{r_2}$

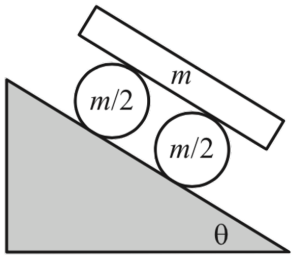
B. $\frac{r_1 - r_2}{r_2}$

C. $\frac{r_1 + r_2}{r_2}$

D. $\frac{r_2 - r_1}{r_2}$

E. $1 - \frac{r_1}{r_2}$

10. Consider the following "car" on an inclined plane.



The system is released from rest, and there is no slipping between any surfaces. Find the ratio of the board's acceleration a_b and the wheels' acceleration a_w.

A. 1

B. $\sqrt{2}/2$

C. $\sqrt{2}$

D. 8/5

E. 2

11. In the setup of the previous problem, find a_b.

A. $4mg\sin\theta/3$

B. $3mg\sin\theta/2$

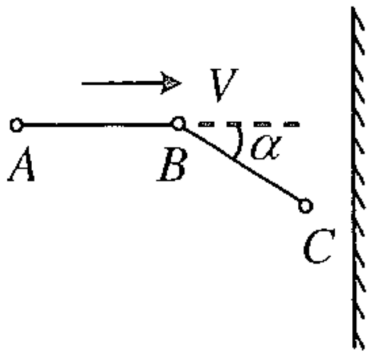
C. $\sqrt{2}mg\sin\theta$

D. $12mg\sin\theta/11$

E. $6mg\sin\theta/5$

12. Three identical small balls, each of mass m, lie on a smooth horizontal table. The balls are smoothly connected by two light rigid rods AB and BC of equal fixed length l, forming an angle $\alpha = 45^\circ$ at point B. Initially, the entire system moves with speed V in the direction of rod AB.

Ball C then collides with a smooth, fixed, vertical wall that is perpendicular to rod AB. The collision between the wall and ball C is completely inelastic, and all other interactions are frictionless.



What is the magnitude of the impulse exerted by the wall on the system?

A. $\frac{6}{5}mV$

B. $\frac{7}{5}mV$

C. $\frac{8}{5}mV$

D. $2mV$

E. $\frac{5}{2}mV$

13. **F=ma 2010 Problem 22** Fictitious forces and effective gravity.

Theory time: Fictitious forces arise when motion is described in a non-inertial reference frame. They do not correspond to physical interactions, but are introduced so that Newton's laws retain their standard form in the accelerating frame.

If the reference frame has translational acceleration $\mathbf{a}_0$ relative to an inertial frame, a particle of mass m experiences a fictitious force

$$\mathbf{F}_{\text{fic}} = -m\mathbf{a}_0.$$

In a rotating frame with angular velocity $\boldsymbol{\omega}$, additional fictitious forces appear:

$$\mathbf{F}_{\text{centrifugal}} = m\,\boldsymbol{\omega}\times(\boldsymbol{\omega}\times\mathbf{r}), \qquad \mathbf{F}_{\text{Coriolis}} = -2m\,\boldsymbol{\omega}\times\mathbf{v}.$$

(To put these forces into practical usage in this problem, take the cross product for angular velocities perpendicular to the plane which position $\mathbf{r}$ rotates in, and you'll obtain the 2-D version!)

When gravity $\mathbf{g}$ acts together with fictitious forces, it is convenient to define an effective gravity,

$$\mathbf{g}_{\text{eff}} = \mathbf{g} - \mathbf{a}_0 + \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \mathbf{r}),$$

excluding the velocity-dependent Coriolis term.

In equilibrium, fluids and suspended objects align with $\mathbf{g}_{\text{eff}}$, and the pressure gradient satisfies

$$\nabla p = \rho\, \mathbf{g}_{\text{eff}}.$$

14. **F=ma 2010 Problem 25** Orbital energies.

Some theory: See if you can figure out what each of the following equations means!

Note: $\langle\rangle$ represent the *time average*, a is defined as the *semi-major axis* of an eclipse.

$$U(r) = -\frac{GMm}{r}$$

$$v^2 = GM\left(\frac{2}{r} - \frac{1}{a}\right)$$

$$K(r) = \frac{1}{2}mv^2 = \frac{GMm}{2}\left(\frac{2}{r} - \frac{1}{a}\right)$$

$$E = K + U = -\frac{GMm}{2a}$$

$$v_{\text{circ}} = \sqrt{\frac{GM}{r}}$$

$$E_{\text{circ}} = -\frac{GMm}{2r}$$

$$\langle K \rangle = \frac{GMm}{2a}$$

$$\langle U \rangle = -\frac{GMm}{a}$$

$$\left\langle \frac{1}{r} \right\rangle = \left\langle \frac{1}{a} \right\rangle$$